

Digital Evidence Management System (DEMS)

System Architecture, End-to-End Workflows & Complete 49 REST API Reference Directory

1. Executive Overview & Technology Stack

The Digital Evidence Management System (DEMS) is an enterprise-grade, legally auditable Spring Boot 3 web application built for law enforcement, forensic laboratories, and judicial authorities. DEMS provides end-to-end evidence tracking, automated SHA-256 cryptographic tamper detection, immutable chain of custody recording, physical QR code barcode generation, and read-only executive analytics.

Component Layer	Technologies & Libraries	Key Responsibilities
Core Framework	Java 21 LTS Spring Boot 3.3.5	Application container, dependency injection, runtime environment
Security & Auth	Spring Security JJWT 0.12.6	Stateless JWT bearer authentication & Role-Based Access Control
Data Engine	Spring Data JPA PostgreSQL / H2	Relational ORM data persistence, JPQL aggregation queries
Barcode & QR	ZXing Core 3.5.3 ZXing JavaSE 3.5.3	Generates 250x250 PNG QR codes saved under uploads/qr/
File Storage	Pluggable Local Storage	Saves evidence files under uploads/cases/{caseNumber}/
Documentation	Springdoc OpenAPI 2.6.0	Interactive Swagger UI API documentation and schemas

2. System Components & Feature Modules

DEMS follows a clean **Feature-Oriented Package Layout** (com.dems.<feature>) paired with a strict layered structure (Controller → Service → Repository/Storage → Database/Disk).

Module Name	Package Path	APIs	Core Features & Responsibilities
Authentication & Users	com.dems.auth com.dems.user	6	User onboarding, roles (ADMIN, POLICE_OFFICER, FORENSIC_EXPERT, COURT_OFFICIAL), JWT token issuance, admin auto-seeding.
Case Management	com.dems.cases	8	Case lifecycle state machine (OPEN → UNDER_INVESTIGATION → CLOSED → ARCHIVED), officer assignment, search.
Evidence Management	com.dems.evidence com.dems.storage	8	Multipart file upload, dual MIME & extension validation, storage engine orchestration, streaming downloads, search.
Integrity & Tamper	com.dems.integrity	3	8KB streaming SHA-256 hash generation, live integrity verification against stored hash, forensic history logging.
Chain of Custody	com.dems.custody	6	Legally auditable immutable custody transfers (PENDING → ACCEPTED / REJECTED), single pending transfer rule.
Audit Logging	com.dems.audit	6	Permanent immutable audit logging for all business actions and unexpected failures (IP, User-Agent, Correlation ID, diffs).

Dashboard & Analytics	com.dems.dashboard	8	Read-only executive summary metrics, module KPI breakdowns, date-range trend aggregations, top active users, system health.
QR Code & Barcode	com.dems.qr	4	ZXing 250x250 PNG QR generation, local image storage under uploads/qr/, ADMIN regeneration, barcode scan resolution.
TOTAL SYSTEM APIs	9 Controllers	49	Complete production REST API suite

3. End-to-End API Workflows

Workflow A: User Authentication & JWT Issuance

1. Client sends POST /auth/login containing email & password.
2. AuthServiceImpl verifies BCrypt password and active account status.
3. JwtService generates a signed HMAC-SHA256 JWT bearer token returned in LoginResponse.

Workflow B: Evidence Upload, Hashing & Auto-QR Generation

1. Client sends multipart POST /evidence/upload with file and metadata.
2. EvidenceValidationService enforces dual MIME-type and extension compatibility.
3. LocalStorageServiceImpl writes file to disk at uploads/cases/{caseNumber}/{uuid}.ext.
4. EvidenceIntegrityServiceImpl streams file bytes to compute initial SHA-256 hash.
5. QRCodeServiceImpl generates 250x250 PNG QR image at uploads/qr/QR-{evidenceNumber}.png.
6. AuditService records an immutable UPLOAD audit event.

Workflow C: Chain of Custody Transfer Handshake

1. Current custodian calls POST /custody/transfer (enforcing single pending transfer rule).
2. Recipient calls POST /custody/{id}/accept.
3. System updates EvidenceEntity.currentCustodian and logs permanent audit events.

Workflow D: Physical Evidence Barcode Scanning & Resolution

1. Barcode scanner hits GET /qr/resolve/{evidenceNumber}.
2. System returns safe QRResolveResponse payload (evidenceNumber, caseNumber, name, status, custodian, integrityStatus).
3. Sensitive file paths, hashes, remarks, and audit logs are **strictly omitted**.

4. Complete Directory of All 49 REST APIs

#	HTTP & Path	Access Role	Function / Description
1	POST /auth/login	Public	Authenticate user credentials and issue signed JWT token
2	POST /users	ADMIN	Create a new system user account
3	GET /users	ADMIN	List users with pagination and search filters
4	GET /users/{id}	ADMIN	Retrieve specific user profile metadata
5	PUT /users/{id}	ADMIN	Update user profile attributes
6	PATCH /users/{id}/status	ADMIN	Activate or disable a user account
7	POST /cases	ADMIN	Create a new investigation case record
8	GET /cases	Authenticated	List all cases with pagination
9	GET /cases/my-cases	Authenticated	List cases assigned to current officer
10	GET /cases/{id}	Authenticated	Retrieve case record details
11	PUT /cases/{id}	ADMIN / Officer	Update case summary and location details
12	PATCH /cases/{id}/status	ADMIN / Officer	Transition case status via CaseStateMachine
13	PATCH /cases/{id}/assign-officer	ADMIN	Assign primary investigating officer to case
14	GET /cases/search	Authenticated	Multi-field dynamic specification search for cases
15	POST /evidence/upload	Authenticated	Upload evidence digital file & metadata
16	GET /evidence	Authenticated	List evidence records with pagination
17	GET /evidence/{id}	Authenticated	Retrieve evidence metadata by ID
18	GET /evidence/download/{id}	Authenticated	Stream raw evidence digital file download
19	PUT /evidence/{id}	Authenticated	Update evidence metadata record
20	PATCH /evidence/{id}/status	Authenticated	Transition evidence lifecycle status
21	GET /evidence/case/{caseId}	Authenticated	List all evidence attached to specific case
22	GET /evidence/search	Authenticated	Multi-field dynamic search for evidence
23	POST /evidence/{id}/verify	Authenticated	Run live SHA-256 check against stored hash
24	GET /evidence/{id}/integrity	Authenticated	Retrieve forensic verification history for evidence
25	GET /evidence/integrity/report	ADMIN / FORENSIC	Retrieve system-wide integrity summary report
26	POST /custody/transfer	Authenticated	Initiate evidence custody transfer request
27	POST /custody/{id}/accept	Authenticated	Accept pending custody transfer request
28	GET /custody/{id}	Authenticated	Retrieve specific custody record details
29	GET /custody/evidence/{evidenceId}	Authenticated	Retrieve active custody status for evidence
30	GET /custody/history/{evidenceId}	Authenticated	Retrieve complete immutable custody timeline
31	GET /custody/search	Authenticated	Search custody records by filters
32	GET /audit	ADMIN	Query immutable system audit log history
33	GET /audit/{id}	ADMIN	Retrieve specific audit log entry details
34	GET /audit/search	ADMIN	Multi-field dynamic specification search for audit logs

35	GET /audit/dashboard	ADMIN	Operational audit action breakdown summary
36	GET /audit/user/{userId}	ADMIN	Retrieve audit activity history by user ID
37	GET /audit/entity/{entityType}/{entityId}	ADMIN	Retrieve audit history for specific entity
38	GET /dashboard/summary	ADMIN / FORENSIC	High-level executive KPI summary metrics
39	GET /dashboard/cases	ADMIN / FORENSIC	Case distributions & monthly creation trends
40	GET /dashboard/evidence	ADMIN / FORENSIC	Evidence file types, largest files, and upload trends
41	GET /dashboard/integrity	ADMIN / FORENSIC	Verification ratios and recent verifications feed
42	GET /dashboard/custody	ADMIN / FORENSIC	Transfer status ratios & average transfer time
43	GET /dashboard/audit	ADMIN / FORENSIC	Audit action breakdowns & top active users ranking
44	GET /dashboard/recent-activities	ADMIN / FORENSIC	Real-time cross-module consolidated activity stream
45	GET /dashboard/system-health	ADMIN / FORENSIC	Operational storage usage, DB status, active users
46	GET /qr/evidence/{evidenceId}	Authenticated	Retrieve QR code metadata and qrDownloadUrl
47	GET /qr/evidence/{evidenceId}/image	Authenticated	Stream 250x250 PNG QR code barcode image
48	POST /qr/evidence/{evidenceId}/regenerate	ADMIN	Regenerate physical QR code PNG image on disk
49	GET /qr/resolve/{evidenceNumber}	Authenticated	Resolve barcode scan to safe, non-sensitive metadata